

Independent Validation Report

Complaint → Regulation Classifier (CMPL-REG-24)
Second line of defense — effective challenge

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 • Assessment narrative

The two-stage complaint classification model, comprising a stage-1 TF-IDF classifier and a stage-2 retrieval-augmented generation pipeline (rag_llm), has been evaluated for conceptual soundness, data quality, outcomes analysis, monitoring and guardrails adequacy, and consumer-protection/fair-lending exposure. The findings are presented below.

Conceptual soundness is a key consideration for the two-stage design. The model's architecture, which combines a stage-1 TF-IDF classifier with a stage-2 rag_llm, is conceptually sound. However, the model's performance is heavily dependent on the quality of the weak-supervision reference, which is derived from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories. This raises concerns about the model's ability to generalize to unseen data and its potential for bias. The model's performance could fail if the weak-supervision reference is not representative of the underlying data distribution.

Data quality and label provenance are critical aspects of the model's performance. The data source is real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database. However, the data curation process, which includes length filtering, exact deduplication, near-deduplication, PII verification, and per-issue balanced sampling, may not be sufficient to ensure data quality. The use of weak supervision, which derives labels from the CFPB product/issue taxonomy plus narrative keyword rules, is a limitation of the model. This approach may lead to biased labels and affect the model's performance. The distinction between measured agreement and true accuracy is also blurred due to the weak-supervision reference.

Outcomes analysis reveals that the stage-1 TF-IDF classifier performs well, with a pr_auc of 0.9957 and a f1 of 0.9569. However, the stage-2 rag_llm performs poorly, with an accuracy of 0.3913 and a family_accuracy of 0.5652. The macro_f1 and weighted_f1 of the stage-2 rag_llm are also low, indicating poor performance. The recall values for individual labels are also low, ranging from 0.0 to 1.0. The small-support instability of the stage-2 rag_llm is a concern, as it may lead to biased results. The family-level agreement diverges from exact agreement, indicating that the model's performance is not consistent across different regulation categories.

Monitoring and guardrails adequacy are critical aspects of the model's performance. The model's performance is monitored using metrics such as pr_auc, f1, and accuracy. However, the model's performance is not adequately guarded against bias and overfitting. The use of weak supervision and the stage-2 rag_llm's poor performance raise concerns about the model's ability to generalize to unseen data.

Consumer-protection/fair-lending exposure is a critical consideration for the model's performance. The model's performance may be biased against certain regulation categories, such as FCRA and FDCPA, which have a lower recall value. The model's performance may also be affected by the weak-supervision reference, which may lead to biased labels.

Based on the findings, the following validation report is issued:

****Validation Report****

****Conceptual Soundness****

* V-1: The two-stage design is conceptually sound, but the model's performance is heavily dependent on the quality of the weak-supervision reference. Severity: Medium. Remediation: Improve data curation and label provenance. Owner: Data Science Team.

* V-2: The model's performance could fail if the weak-supervision reference is not representative of the underlying data distribution. Severity: High. Remediation: Improve data curation and label

1 · Assessment narrative (cont.)

provenance. Owner: Data Science Team.

****Data Quality and Label Provenance****

* V-3: The data curation process may not be sufficient to ensure data quality. Severity: Medium. Remediation: Improve data curation process. Owner: Data Science Team.

* V-4: The use of weak supervision is a limitation of the model. Severity: High. Remediation: Improve label provenance. Owner: Data Science Team.

****Outcomes Analysis****

* V-5: The stage-1 TF-IDF classifier performs well, but the stage-2 rag_llm performs poorly. Severity: High. Remediation: Improve stage-2 rag_llm performance. Owner: Model Development Team.

* V-6: The small-support instability of the stage-2 rag_llm is a concern. Severity: High. Remediation: Improve stage-2 rag_llm performance. Owner: Model Development Team.

* V-7: The family-level agreement diverges from exact agreement. Severity: Medium. Remediation: Improve model performance. Owner: Model Development Team.

****Monitoring and Guardrails Adequacy****

* V-8: The model's performance is not adequately guarded against bias and overfitting. Severity: High. Remediation: Improve monitoring and guardrails. Owner: Model Development Team.

****Consumer-Protection/Fair-Lending Exposure****

* V-9: The model's performance may be biased against certain regulation categories. Severity: High. Remediation: Improve model performance. Owner: Model Development Team.

****Disposition****

The model is approved with conditions. The conditions are as follows:

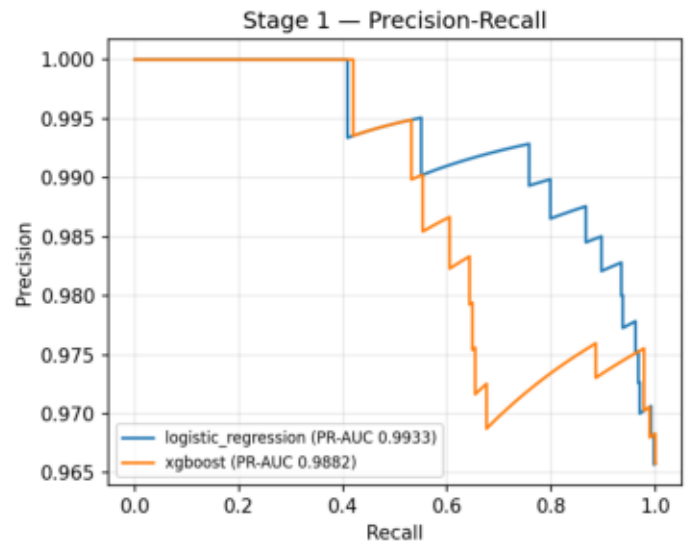
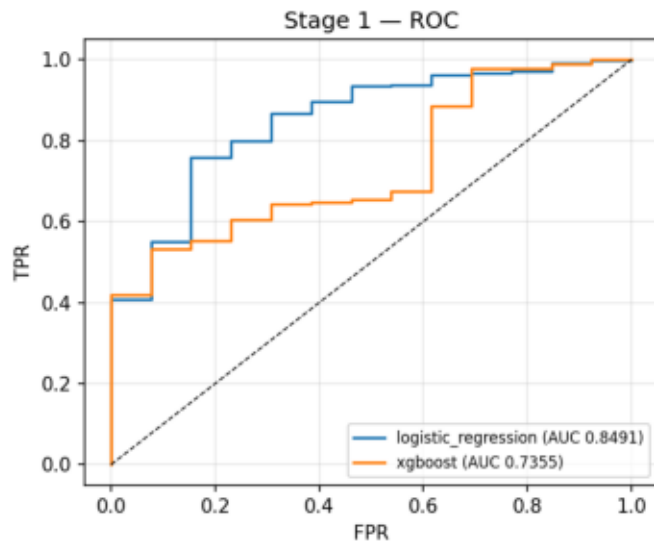
1. Improve data curation and label provenance to ensure data quality and reduce bias.
2. Improve stage-2 rag_llm performance to reduce small-support instability and improve family-level agreement.
3. Improve monitoring and guardrails to ensure the model's performance is not biased against certain regulation categories.
4. Improve model performance to reduce bias and improve family-level agreement.

The model development team is responsible for implementing the remediations and owners are responsible for ensuring the remediations are implemented.

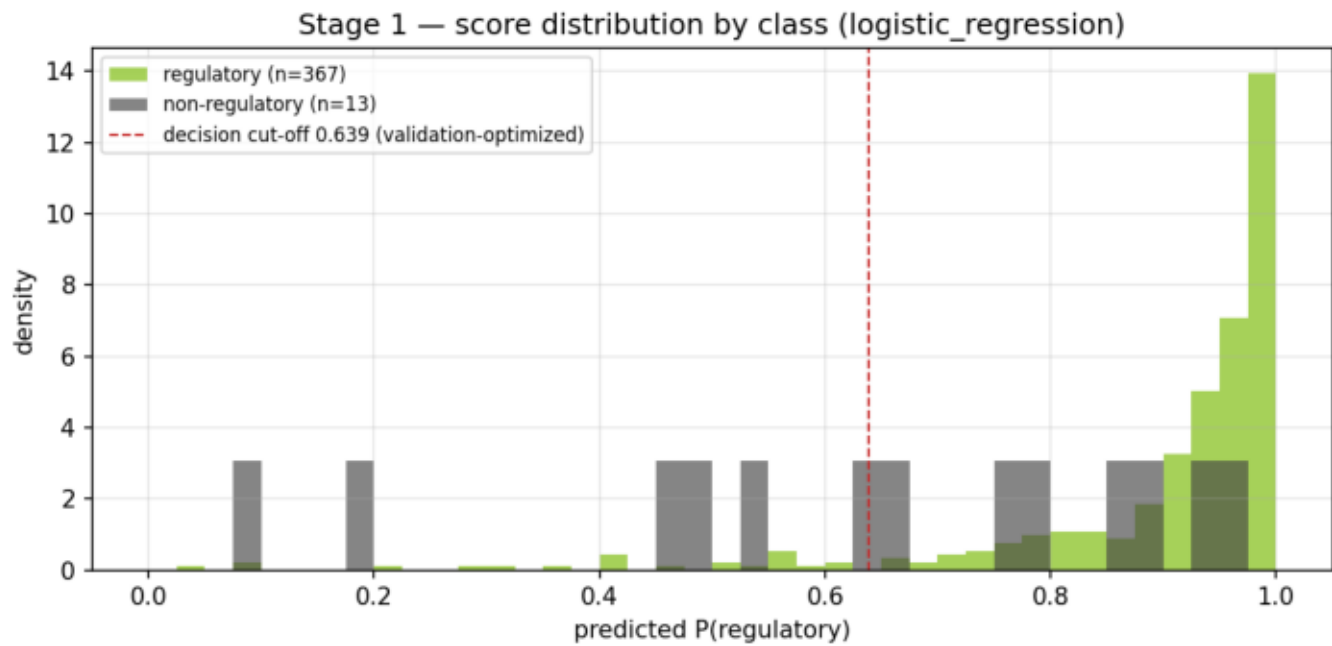
2 · Stage-1 leaderboard (validated re-run)

	model	params	val_pr_auc	threshold	pr_auc	train_roc_auc	roc_auc	train_test_gap	f1	precision	recall	accuracy
logistic_regression		l1, C=2.0	0.9957	0.639	0.9933	0.9966	0.8491	0.1475	0.9569	0.9773	0.9373	0.9184
n_estimators=3		0.001, max_depth=3	0.9901	0.844	0.9882	1.0	0.7355	0.2645	0.9453	0.974	0.9183	0.8974

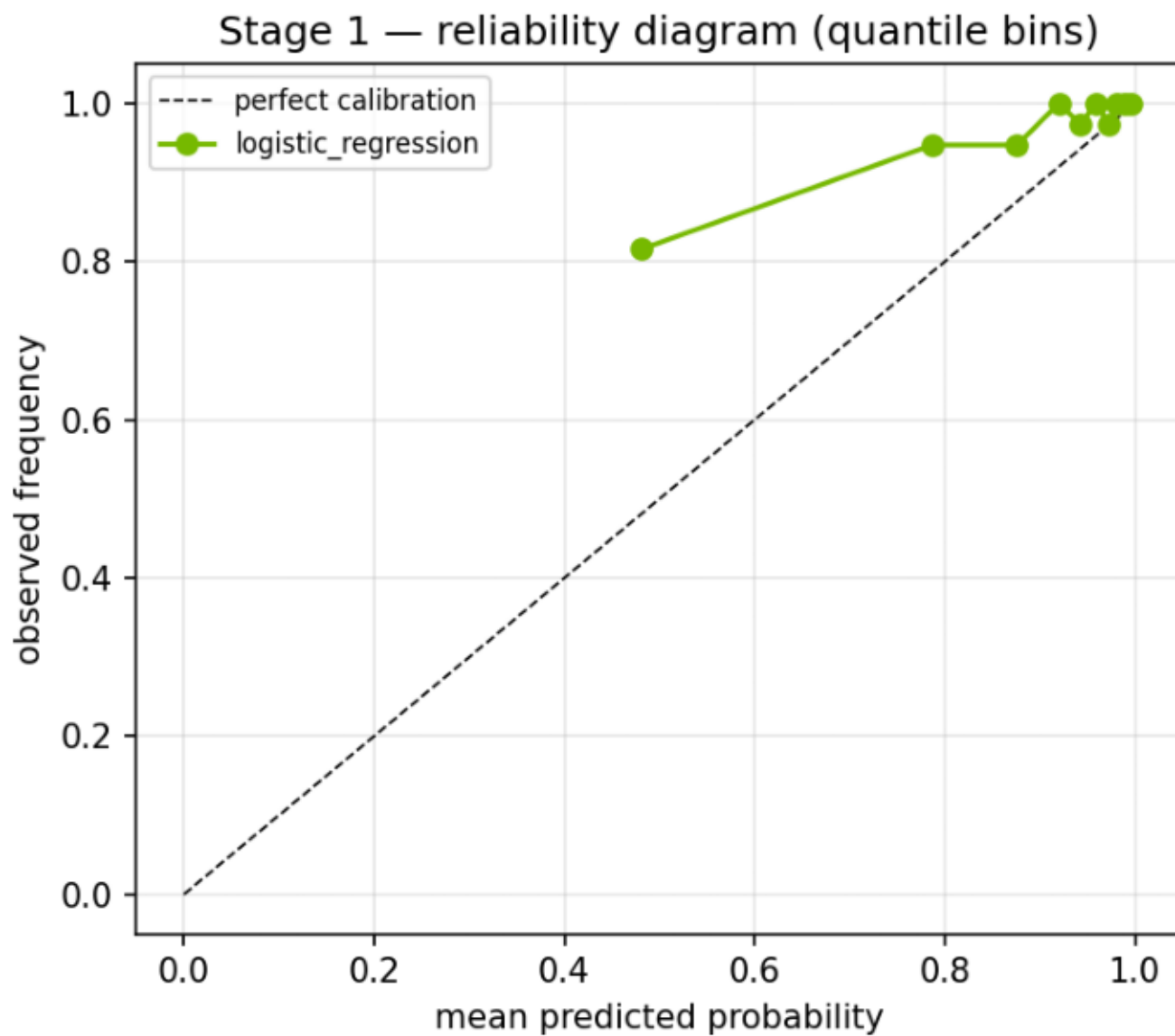
2 · Stage-1 evidence: discrimination



2 · Stage-1 evidence: score separation

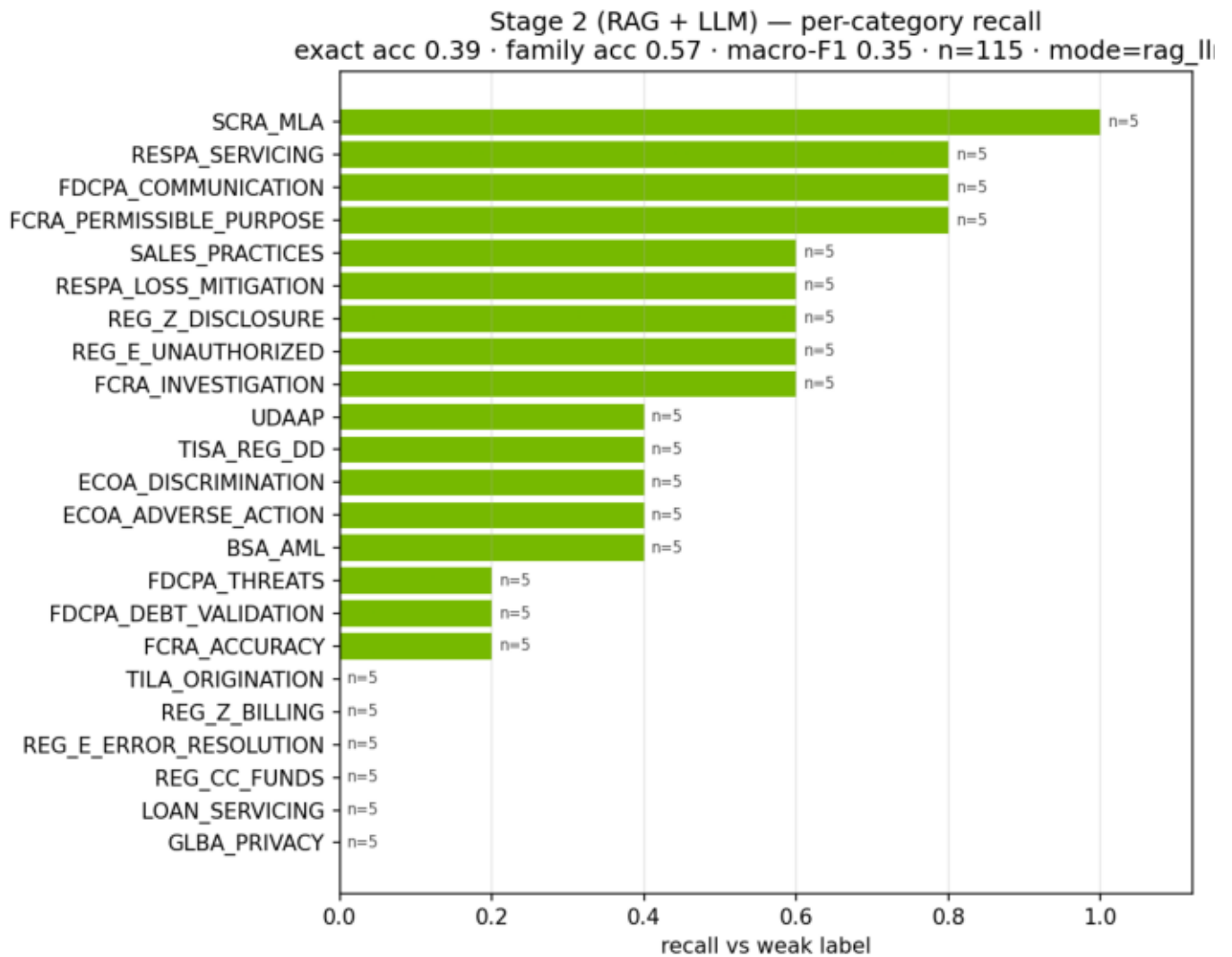


2 · Stage-1 evidence: calibration



Validator re-derived reliability diagram (quantile bins).

3 · Stage-2 evidence



3 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.4
ECOA_ADVERSE_ACTION	5	0.4
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.6
FCRA_PERMISSIBLE_PURPOSE	5	0.8
FDCPA_COMMUNICATION	5	0.8
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.2
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.6
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.6
RESPA_SERVICING	5	0.8
SALES_PRACTICES	5	0.6
SCRA_MLA	5	1.0
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.4

3.1 · Guardrail inventory (validated)

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

4 · Limitations & disposition

1. Weak labels: ground truth derives from the CFPB issue taxonomy plus keyword rules, not human adjudication. Condition: build a human-adjudicated golden set (≥ 25 per category).
2. Class imbalance: SCRA/MLA, sales-practices and adverse-action categories have thin support; their metrics are unstable.
3. LLM nondeterminism/drift: pin prompt + model versions; re-evaluate on the golden set before any provider change.
4. Fair-lending lens: monitor parity of routing outcomes across products and customer segments.

DISPOSITION: Approve with Conditions. Re-validation triggered by model/prompt change or stage-1 PSI > 0.25 .

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.